



UNIVERSITI MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSE3433
Software Architecture

Cloud-Native Learning Management System (LMS)
(Cloud-Native / Layered Architecture)

Prepared by:
Group No.5

No.	Name	Matric No.
1	NUR MAS QASRINA BINTI MAS JAYA	S74706
2	AIDA HUSNA BINTI MUHAMAD ZAIDI	S76103
3	NURSYAHEERA BINTI MOHD HISHAM	S76276

Prepared for:
DR MOHD NOR HASSAN

COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOUR
SEMESTER II 2025/2026

WEEKLY LOG ENTRIES

PHASE 1: Project Proposal

WEEK 5

Undertaken Tasks & Weekly Activities	● Participated in group discussions to identify problems that faced by traditional LMS platforms. ● Chooosed the role for each member. ● Discussed some suitable technologies and database solutions for the proposed cloud-native LMS. ● Assisted in preparing the Project Proposal document for Deliverable 1.
Individual Contributions (as Backend & Database Engineer)	● Suggested the use of RESTful APIs to enable communication between system modules because Cloud-Native LMS consists of multiple independent services. ● Proposed suitable databases such as MongoDB and SQLite for managing LMS data. ● Identified the data requirements for user management, course management, assignment and progress tracking modules. ● Shared ideas on how backend services would interact with the database layer. ● Shared ideas on backend requirements for user authentication and assignment submission modules.
Challenges Encountered	Limit understanding of how database per service concepts work in cloud-native architecture as of how backend services interact with cloud-native environment.
Action Plans	Studied and reviewed some examples of cloud-native LMS implementation and REST API architecture to enhance the understanding.

Table 1 : Weekly Progress Log (Week5)

PHASE 2: Architecture Design

WEEK 6

Undertaken Tasks & Weekly Activities	● Participated in group discussion to analyse the function and non-function requirement of the Cloud-Native LMS. ● Reviewed the proposed architecture style and identified the responsibilities of backend services within the Application Layer. ● Discussed how the frontend, backend services, API Gateway and database would communicate with each other. ● Explore some suitable approaches for implementing RESTful APIs to support LMS functionalities such as user management, assignment submissions and progress tracking.
---	--

	- Identified the data requirement needed by each module and how these modules would interact with database layer. - Assisted in refining the system overview and ensuring that backend-related components aligned with the proposed architecture design.
Individual Contributions (as Backend & Database Engineer)	- Identified backend responsibilities for each microservice or layer including User Management, Assignment Service and Notification Service. - Discussed the finding about the database integration requirement for the proposed system by identifying how backend services would connect and interact with the database layer. - Suggested using RESTful APIs to provide standardized communication between system components. - Reviewed the proposed service interactions to ensure the backend design supported scalability and maintainability.
Challenges Encountered	- Gained insight into efficient communication methods between multiple microservices while maintaining loose coupling principles. - Identified optimal strategies for organizing backend services to prevent redundant responsibilities among system modules. - Developed a scalable database design capable of supporting future growth and higher user traffic without structural overhauls. - Evaluated the trade-offs between system complexity and long-term maintainability regarding the number of independent services.
Action Plans	- Investigate advanced microservices interaction models and optimal RESTful API design standards. - Analyze existing cloud-native LMS frameworks to gain technical clarity on backend deployment methods. - Update the data schema and component duties following group reviews and supervisor suggestions. - Map out API endpoints and information routing to streamline the upcoming prototyping process. - Deepen technical proficiency in Spring Boot or Node.js and database connection logic for implementation readiness.

Table 2 : Weekly Progress Log (Week 6)

WEEK 7

Undertaken Tasks & Weekly Activities	Conducted a comprehensive review of the finalized architectural diagrams and technical documentation prior to the Deliverable 2 submission.
---	---

	● Engaged in group consultative sessions to validate the alignment of the proposed backend framework with established system specifications. ● Verified the visual representation of workflows between frontend modules, backend logic, and the persistent storage layer within the diagrams. ● Collaborated with team members to detect and rectify any technical discrepancies or omissions in the interaction and deployment models. ● Participated in the final assessment of the architecture report to confirm that every service facilitates the core functionalities of the LMS.
Individual Contributions (as Backend Developer)	● Reviewed the sequence diagrams involving backend processes such as user registration, assignment submission, notifications, and progress tracking. ● Verified that the proposed backend workflows supported the functional requirements specified in the report. ● Ensured that the responsibilities assigned to each microservice were clearly defined without overlapping functionalities. ● Examined whether database interactions illustrated in the diagrams reflected the intended flow of data between the application and data layers. ● Assisted in proofreading backend-related sections of the report to maintain consistency and technical accuracy.
Challenges Encountered	● It was challenging to ensure that all sequence diagrams accurately reflected the actual flow of backend operations while remaining simple and easy to understand. ● Some service responsibilities appeared to overlap, making it difficult to determine the appropriate boundaries between microservices. ● Translating conceptual backend processes into UML diagrams without omitting important details required multiple revisions. ● Maintaining consistency between the functional requirements and the proposed service interactions was time-consuming.
Action Plans	● Revisit the sequence diagrams and service descriptions to improve clarity and accuracy. ● Refine the boundaries of each microservice to avoid duplicated responsibilities. ● Use the finalized diagrams as references during the prototype development phase to ensure the implementation follows the proposed architecture. ● Document the expected backend processes and API interactions to simplify future development activities.

	● Prepare for prototype implementation by reviewing backend technologies and database integration approaches identified during the design phase.
--	--

Table 3 : Weekly Progress Log (Week 7)

PHASE 3: Project Observation 1 & Prototype Progress WEEK 8

Undertaken Tasks & Weekly Activities	● Participated in group discussions to prepare for Project Observation 1. ● Evaluated the architectural visualisations and designated essential backend modules for initial focus in the upcoming prototyping stage. ● Started planning the implementation approach for backend services required by the LMS. ● Discussed database structures and the data flow needed to support the core functionalities of the system. ● Prepared references and notes related to backend development to facilitate the transition from design to implementation.
Individual Contributions (as Backend Developer)	● Identified backend functionalities that should be implemented first such as user registration, login authentication and assignment submission. ● Planned the database tables required to support the prototype, including user, course, assignment, notification and progress tracking records. ● Drafted the expected CRUD operations that each backend service would perform. ● Reviewed how RESTful APIs could be structured to support communication between the frontend and backend components. ● Assisted the team in preparing explanations regarding backend processes for Project Observation 1.
Challenges Encountered	● Determining which backend functionalities should be prioritised for the initial prototype due to time constraints. ● Translating the proposed architecture into practical implementation steps while ensuring consistency with the original design. ● Understanding how multiple modules would share and retrieve data efficiently without affecting system performance. ● Ensuring that the planned database structure was flexible enough to accommodate future enhancements.

Action Plans	• Determine the order of priority for executing core backend modules, focusing on login security and assignment management. • Complete the database architecture design and establish the necessary query structures for the upcoming coding phase. • Subdivide backend engineering goals into manageable phases to ensure a structured and efficient development process. • Maintain ongoing research into industry standards for API design and seamless data persistence integration. • Refine the approach for building the initial system model by incorporating critiques gathered from the first observation session.
---------------------	--

Table 4 : Weekly Progress Log (Week 8)

Tool Used	ChatGPT
Purpose	Assisting in planning the implementation of backend services for the LMS prototype.
Prompt Used	"What backend functionalities should be prioritised when developing the first prototype of a cloud-native learning management system?"
Output Summary	Suggested prioritising core functionalities such as user authentication, course management, assignment submission, and database integration before implementing advanced features.
How It Was Used	The recommendations were used as a guideline to determine the priority of backend functionalities during the early prototype stage. Based on the suggestions, I focused on planning essential components such as user authentication, assignment submission, and database integration before considering additional features. This helped the team prepare a clearer explanation of the prototype scope and backend workflow for Project Observation 1.

Table 5 : GenAI Usage Record (Week 8)

WEEK 9

Undertaken Tasks & Weekly Activities	• Participated in Project Observation 1 to present the proposed Cloud-Native LMS architecture and prototype planning. • Explained the backend design and how the Application Layer would support the system functionalities. • Recorded feedback and recommendations provided by the lecturer regarding the proposed architecture. • Discussed improvements required before proceeding with further prototype development.
---	---

	Reviewed the feasibility of the planned backend implementation based on the feedback received.
Individual Contributions (as Backend Developer)	- Elaborated on the architectural role of backend services in managing client requests and facilitating data exchange with the persistence layer. - Provided technical clarity on the utilization of RESTful APIs to ensure seamless communication between frontend modules and backend logic. - Presented the conceptual framework for secure user registration and authentication workflows within the system. - Evaluated the proposed database schema to verify its capacity to accommodate core LMS functionalities and various user permissions. - Refined backend security protocols and storage management strategies following the analytical review and feedback from the supervisor.
Challenges Encountered	- Explaining technical backend concepts in a way that was easy to understand during the observation session. - Addressing questions regarding data security and how sensitive user information would be protected. - Ensuring that the proposed backend design remained scalable while still being practical for prototype development. - Interpreting the lecturer's feedback and determining which improvements should be prioritised.
Action Plans	- Refine the authentication workflow by incorporating appropriate security practices such as password hashing and role-based access control. - Review the database design to ensure user information and academic records are managed efficiently. - Prioritise the implementation of improvements highlighted during Project Observation 1. - Prepare a clearer backend implementation plan to guide the next phase of prototype development. - Continue strengthening knowledge of backend security and API development practices before implementation.

Table 6 : Weekly Progress Log (Week 9)

Tool Used	Gemini
Purpose	Strengthened technical knowledge regarding backend security protocols

	and industry standards for safeguarding confidential user data within the LMS.
Prompt Used	"What are the recommended practices for securing user authentication and storing user credentials in a learning management system?"
Output Summary	The system suggested adopting cryptographic password hashing, enforcing role-based access restrictions, and implementing secure session handling while limiting data exposure for private user details.
How It Was Used	The recommendations were used to review the proposed authentication workflow and identify potential improvements to the backend design. The suggested security practices helped me understand how user credentials should be protected and how access permissions could be managed according to user roles. This information was also considered when discussing enhancements to the LMS after receiving feedback during Project Observation 1.

Table 7: GenAI Usage Record (Week 9)

WEEK 10

Undertaken Tasks & Weekly Activities	- Continued the development planning for the LMS prototype based on feedback received during Project Observation 1. - Reviewed the proposed backend workflows and identified areas that required refinement before implementation. - Discussed how database structures would support the functional requirements of the LMS. - Explored suitable approaches for implementing backend logic and handling data exchange between system components. - Worked with team members to ensure that the backend design remained consistent with the architecture documentation.
Individual Contributions (as Backend Developer)	- Established the definitive data architecture by designating primary entities and relational attributes for the LMS services. - Conducted a technical assessment of CRUD operations supporting User Management, Course Management, and academic tracking modules. - Analyzed backend service logic regarding request validation and data processing protocols prior to persistence layer interaction. - Validated the alignment of planned backend workflows with the functional system specifications established in the project proposal. - Assisted in the formal documentation of backend operational flows to facilitate technical guidance for future implementation phases.
Challenges	Determining the most appropriate database structure to support

Encountered	multiple LMS functionalities without introducing unnecessary complexity. • Ensuring that all backend processes remained aligned with the proposed architecture and functional requirements. • Identifying potential issues in data flow between modules before actual implementation. • Managing time effectively while preparing backend documentation and planning future development activities.
Action Plans	• Refine the database schema and validate relationships between entities to improve data consistency. • Prepare a list of API endpoints and expected request-response structures for each backend service. • Review additional references on backend implementation techniques and best practices. • Continue coordinating with team members to ensure smooth integration between frontend and backend components. • Use the documented workflows as a guide during the implementation and testing phases.

Table 8 : Weekly Progress Log (Week 10)

Tool Used	ChatGPT
Purpose	Strengthened technical knowledge regarding the conceptualization of a scalable data architecture and the establishment of technical specifications for RESTful API integration within the LMS framework.
Prompt Used	"How should I design a database schema and RESTful API structure for a cloud-native learning management system?"
Output Summary	The system recommended the identification of primary entities including users, courses, assignments, submissions, notifications, and academic records. Furthermore, it offered technical illustrations for RESTful API endpoints and detailed the integration of CRUD operations across the various LMS modules.
How It Was Used	The recommendations were used as a reference when reviewing the proposed database structure and identifying the API operations required for each LMS module. The examples provided helped me visualise how backend services would process requests and interact with the database. This improved the clarity of the backend documentation and provided a better understanding of the implementation approach for the prototype.

Table 9: GenAI Usage Record (Week 10)

WEEK 11

Undertaken Tasks & Weekly Activities	• Continued reviewing the LMS prototype progress to ensure that the proposed backend design was practical and achievable. • Evaluated whether the planned backend workflows supported the functional requirements specified in the project documentation. • Discussed potential testing scenarios for core LMS functionalities such as user authentication and assignment submission. • Reviewed the interaction between backend services and the database layer to identify possible issues before implementation. • Assisted the team in preparing updates on prototype progress and documenting improvements made after Project Observation 1.
Individual Contributions (as Backend Developer)	• Conducted a technical review of the architectural logic governing User Management, Assignment Submission, and Progress Tracking modules. • Validated the alignment of planned CRUD operations with the functional system specifications established for each LMS module. • Assessed the capacity of the database schema to manage complex data relationships and diverse user access permissions. • Determined essential validation protocols for backend logic, including the prevention of redundant registrations and enforcement of submission deadlines. • Assisted in the formal documentation of backend testing scenarios and anticipated system responses during operational execution.
Challenges Encountered	• Predicting potential technical hurdles within the backend services in the absence of a comprehensive testing environment. • Confirming that the architectural planning accounts for all necessary business logic and data validation protocols. • Evaluating edge cases, including erroneous user inputs and overdue task submissions, that could impact system usability. • Aligning the structural design and functional system specifications with the anticipated goals of the initial prototype.
Action Plans	• Develop a checklist of backend validation rules and expected outcomes for future testing activities. • Refine the planned business logic for critical LMS functions, particularly authentication and assignment handling. • Continue reviewing database relationships to minimise data inconsistencies. • Prepare backend documentation that can guide implementation and testing during the next project phase. • Incorporate any additional improvements before Project Observation 2.

Table 10 : Weekly Progress Log (Week 11)

Tool Used	ChatGPT
Purpose	To understand common backend validation techniques and testing considerations for LMS functionalities.
Prompt Used	"What validation rules and testing scenarios should be considered when developing backend functionalities for a learning management system?"
Output Summary	The system output recommended the establishment of rigorous validation protocols, including the mitigation of redundant user registrations, the verification of mandatory input parameters, and the strict enforcement of submission temporal constraints. Furthermore, it emphasized implementing role-based access restrictions and conducting comprehensive scenario testing to validate the operational integrity of the backend framework.
How It Was Used	The recommendations were used to identify the validation rules and expected behaviours that should be considered for the LMS backend. The suggested testing scenarios helped me think about how different user actions, such as invalid logins, duplicate registrations, and late assignment submissions, should be handled by the system. These insights were incorporated into the backend review process to ensure that the proposed workflows aligned with the functional requirements and provided a better user experience.

Table 11: GenAI Usage Record (Week 11)

PHASE 4: Project Observation

WEEK 12

Undertaken Tasks & Weekly Activities	• Participated in Project Observation 2 to present the updated progress of the Cloud-Native LMS project. • Reviewed the improvements made to the proposed backend design following feedback from Project Observation 1. • Discussed the readiness of the backend components and database design for the final phase of the project. • Evaluated whether the planned backend workflows aligned with the functional and non-functional requirements of the LMS. • Recorded additional feedback and recommendations provided during the observation session for further refinement.
Individual Contributions (as Backend Developer)	• Detailed the architectural refinements made to backend services and persistent storage models following the integration of prior consultative feedback.

	• Elucidated the implementation of RESTful API protocols to facilitate standardized data exchange between the presentation layer and system logic modules. • Conducted a technical verification of the data schema to confirm support for complex entities such as user permissions, task submissions, and academic tracking. • Provided technical clarity on how validation logic and secure authentication workflows enhance the operational integrity and protection of user credentials. • Collaborated in the analytical review of the backend framework to isolate and resolve any technical hurdles prior to final documentation completion.
Challenges Encountered	• Explaining the proposed backend improvements clearly despite not having a fully implemented prototype. • Ensuring that all lecturer feedback from the previous observation had been appropriately addressed. • Demonstrating how the planned backend processes would meet both functional and non-functional requirements. • Balancing the level of technical detail during the presentation so that it remained understandable to the audience.
Action Plans	• Finalise backend-related documentation for inclusion in the final report. • Review all backend workflows and database descriptions to ensure consistency across project deliverables. • Address any remaining comments received during Project Observation 2 before the final submission. • Prepare supporting explanations for backend decisions and implementation approaches. • Conduct a final review of all project documents to ensure accuracy and completeness.

Table 12 : Weekly Progress Log (Week 12)

Tool Used	ChatGPT
Purpose	To obtain suggestions for improving backend documentation and ensuring that the proposed design addressed the project requirements effectively.
Prompt Used	"How can I review and improve backend documentation for a cloud-native learning management system before final submission?"
Output Summary	The system output recommended the verification of structural alignment between functional specifications, architectural visualizations, and backend operational flows. Furthermore, it emphasized providing

	technical clarity regarding validation protocols, secure authentication workflows, and the definitive responsibilities designated for every microservice.
How It Was Used	The recommendations were used as a checklist to review the backend sections of the project documentation before Project Observation 2. I compared the functional requirements with the proposed backend workflows and database descriptions to identify any inconsistencies. The suggestions also helped me prepare clearer explanations regarding authentication, validation rules, and service responsibilities during the observation session.

Table 13: GenAI Usage Record (Week 12)

WEEK 13

Undertaken Tasks & Weekly Activities	• Participated in the final review and preparation of the project deliverables before submission. • Reviewed the completed project documentation to ensure consistency between the proposal, architecture design, and final report. • Assisted the team in checking that all functional and non-functional requirements had been adequately addressed. • Verified that the backend-related explanations accurately reflected the proposed Cloud-Native LMS architecture. • Conducted a final proofreading session to identify and correct any errors in the documentation.
Individual Contributions (as Backend Developer)	• Conducted a technical validation of the backend report sections, focusing on service definitions, data persistence models, and inter-module communication protocols. • Verified the structural alignment between documented server-side workflows, functional system specifications, and the finalized architectural visualizations. • Cross-referenced the persistent storage schema and designated service duties to maintain technical consistency across the entire deliverable. • Refined technical explanations concerning secure authentication logic, academic task processing, and student progress monitoring from an engineering perspective. • Contributed to the final quality assurance audit by isolating and rectifying technical discrepancies or ambiguities prior to project submission.
Challenges Encountered	• Ensuring consistency across multiple project deliverables prepared at different stages of the semester. • Identifying minor discrepancies between the proposal,

	architecture documentation, and final report. • Managing time effectively while reviewing extensive documentation within the submission deadline. • Making sure that technical explanations remained clear and understandable without losing important details.
Action Plans	• Submit the finalized project deliverables within the required timeframe. • Archive project documents and references for future academic use. • Reflect on the strengths and limitations of the proposed backend design to identify areas for improvement. • Continue strengthening knowledge in backend development, RESTful API implementation, and database integration for future projects. • Apply the experience and lessons learned from this project to enhance future software engineering practices.

Table 14 : Weekly Progress Log (Week 13)

Tool Used	ChatGPT
Purpose	To improve the clarity and professionalism of backend-related explanations in the final report.
Prompt Used	"How can I improve the clarity and structure of backend documentation in a software architecture report?"
Output Summary	The system output recommended the verification of structural alignment between functional specifications, architectural visualizations, and backend operational flows. Furthermore, it emphasized providing technical clarity regarding validation protocols, secure authentication workflows, and the definitive responsibilities designated for every microservice.
How It Was Used	The recommendations were used to review and refine the backend sections of the final report before submission. I compared the descriptions of backend services, database interactions, and API communication with the architecture diagrams and functional requirements to ensure consistency. The suggestions also helped improve the clarity of technical explanations, making the documentation easier to understand while maintaining its accuracy and professionalism.

Table 15: GenAI Usage Record (Week 13)